HealthExpress
AI-Powered • Personalized • 24/7 Healthcare

Better Care Closer to You

HealthExpress AI

Voice-First Multilingual AI Triage, Real-Time Telehealth, 15-Minute Hyperlocal Medicine Delivery & Government Health Schemes Integration

Smarter Healthcare for Every Citizen

Accessible Affordable Accountable

AI at the Core. Care Everywhere.

A Healthier India, Together

Engineered and Synthesized with IBM Bob





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The Healthcare Crisis vs. The HealthExpress Solution

Solving fragmentation, language barriers, and emergency response latency in regional healthcare

Current Healthcare Bottlenecks

Fragmented Systems
Patients juggle separate apps for doctor consultations, lab tests, pharmacies, and hospital bed searches.

Language & Literacy Barrier
Over 80% of regional patients cannot navigate English-heavy text interfaces.

Emergency Dispatch Delay
Average medicine delivery takes 60–120 minutes, worsening acute clinical conditions.

Disconnected Medical Records
Doctors lack instant access to past medical history and allergies during emergencies.

Unified Golden 15-Minute Loop

Single Super-App Platform
Instant handoff from AI symptom triage to doctor video call, e-prescription, and medicine delivery.

Multilingual Voice-First Ingestion
Speak naturally in Telugu, Hindi, English, Tanglish, or Hinglish with zero typing.

15-Minute Dark Store Network
Geofenced 3–5 km pharmacy dark stores for guaranteed fast-pack doorstep drop.

Dynamic ABDM QR Protocol
15-minute tokenized patient health record unlocks without data hoarding.





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Engineered with IBM Bob

How IBM Bob served as our core AI software development partner across the entire stack

1. Architecture & Schema Synthesis

- Translated complex healthcare requireiments (ABDM QR consent, 15-minute SLAs) into a unified technical topology.
- Generated the complete 16-table normalized MariaDB relational database schema with audit constraints.

2. Polyglot Full-Stack Code Generation

- Concurrently developed Flutter patient/doctor app, React 19 administrative console, and PHP 8.2+ REST backend.
- Maintained architectural consistency across all 3 codebases without specification drift.



IBM Bob
Builds Healthcare
for a Healthier Tomorrow

3. Real-Time AI & Audio Engineering

- Implemented client-side RMS Voice Activity Detection (VAD) algorithms in Web Audio API.
- Integrated Sarvam regional STT/TTS auto-language detection and resolved WebRTC GainNode loopback echoes.

4. Bob Shell CI/CD & Automated Dual Deployment

- Built autonomous PowerShell dual-compilation automation (deploy.ps1) for Flutter Web and React Admin.
- Automated continuous delivery directly to GitHub Pages with zero static analysis errors.





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Full-Stack Technology Matrix

Robust, scalable polyglot architecture engineered for high availability and low latency

LAYER	TECHNOLOGY / PLATFORM	KEY IMPLEMENTATION DETAILS
1 Mobile & Patient Web	Flutter 3.x (Dart)	Riverpod/Provider state, glassmorphic UI, Web Audio API VAD, Mapbox GL
2 Operations Dashboard	React 19 + Vite	12 operational workspaces, TailwindCSS, live telemetry charts, KYC review queues
3 Cloud REST API	PHP 8.2+ (Hostinger Cloud)	16 modular endpoints, Apache mod_rewrite, PDO connection pooling, JWT security
4 Production Database	MariaDB Relational Engine	16 normalized tables, foreign key constraints, spatial indexing, audit logs
5 Multimodal Voice AI	Sarvam AI + NVIDIA NIM	saaras:v3 STT, bulbul:v3 regional TTS, and GPT-OSS-20B clinical SOAP engine
6 Video Telehealth	Agora RTC Engine	Encrypted HD WebRTC video, dynamic room token generation, volume metering
7 Geospatial GPS	Mapbox GL	Vector tiles, live rider GPS coordinates, hospital geofencing
8 Payment Gateway	Razorpay Live	UPI, Cards, NetBanking with HMAC-SHA256 signature verification





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Multi-Role Platform Ecosystem

Role-Based Access Control (RBAC) connecting patients, doctors, hospitals, and dark stores

Patient Super-App
(Flutter Mobile & Web)



- Multilingual voice AI intake and AI Lens symptom scanner.
- In-clinic & Agora HD video appointments with specialists.
- 15-minute medicine ordering with live Mapbox GPS tracking.
- ABDM 15-minute QR consent & Aarogyasri 5L health pass validation.

Doctor Clinical Portal
(Flutter Mobile & Web)



- Video, audio, and in-clinic consultation queues.
- Pre-consultation AI SOAP notes and differential diagnosis review.
- Digital e-prescription builder with auto-dosage guidelines.
- ABDM QR scanner for instant past medical history unlock.

Hospital Command Center
(React 19 Admin)



- Live bed capacity management: General, ICU, and Trauma wards.
- Patient admission, bed allocation, and emergency transfer workflows.
- Medical staff rosters and primary/visiting doctor affiliations.
- Aarogyasri cashless pre-authorizations and insurance claims.

Super Admin & Pharmacy Control
(React 19 Admin)



- Platform-wide KPI telemetry: Active volume, orders, and revenue ledgers.
- Dark-store drug license verification and KYC approval queues.
- Doctor Medical Council registration verification.
- Production REST API health telemetry and database connection diagnostics.





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Hyperlocal 15-Minute Dark Store Logistics

Geofenced pharmacy network, automated picking workflows, and live vector GPS tracking

3–5 km Dark Store Network

Local stock. Faster fulfillment. Healthier communities.

Geofenced Dispatch

Algorithm assigns orders to the closest verified dark store within a 3–5 km radius.

3-Minute Fast-Pack Protocol

Digital picking terminal organizes items by rack location, batch number, and expiry date.

Automated Stock Verification

Real-time inventory synchronization prevents out-of-stock cancellations.

Cold-Chain Management

Priority packaging workflows for insulin and temperature-sensitive biologics.



Mapbox GL Live GPS Engine

Real-time visibility. Accurate ETAs. Safer deliveries.

Real-Time Vector Map Tracking

Turn-by-turn live rider location rendered on interactive Mapbox GL vector tiles.

Accurate Predictive ETA

Route calculation accounting for traffic congestion and building access time.

Verified Delivery Handshake

Secure OTP / QR verification ensures correct medication handover at doorstep.

Rider Telemetry

Super Admin live view monitoring active delivery fleet density and SLA compliance.







Smarter Logistics. Healthier Lives.

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Real-Time Telehealth & Agora WebRTC Engine

Encrypted, low-latency HD video consultations with synchronized in-call clinical tools

WebRTC Video & Audio Infrastructure

POWERED BY agora

Low-Latency Streaming

Sub-200ms end-to-end latency powered by Agora RTC Engine.

Dynamic Token Authentication

Single-use, time-expiring RTC tokens generated via PHP backend tied to validated appointment IDs.

Echo & Loopback Suppression

Web Audio API GainNode routing prevents loopback echoes and feedback on low-end smartphones.

Multi-Platform Support

Seamless WebRTC performance across Flutter Mobile, Flutter Web, and desktop browsers.

Synchronized In-Call Physician Workspace

Live AI SOAP Summary

Physician reviews pre-intake AI notes while speaking to the patient.

ABDM Record Access

Scan patient QR during the call to inspect past lab results and allergy history.

Live Digital E-Prescription

Build, validate dosage limits, and digitally sign prescriptions inside the call UI.

Instant Cart Handoff

Ending the call immediately unlocks the patient's pre-filled 15-minute medicine order.





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AI Vision Lens & Prescription Scanner

Multimodal computer vision for physical symptom analysis and optical prescription parsing

01 Symptom Visual Intake

See. Analyze. Get Guidance.



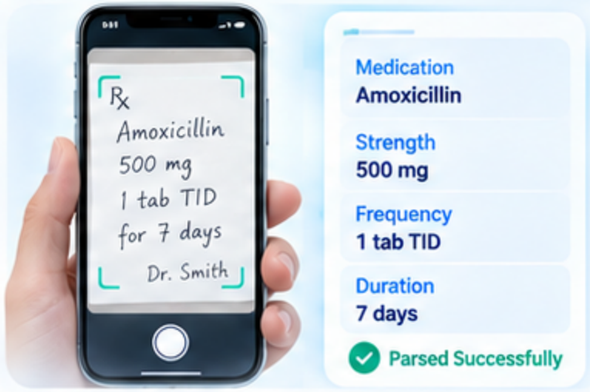
Real-time camera capture
for external skin rashes, wounds, eye irritations, and physical injuries.

Instant classification
of visual severity and potential contagion markers.

Guidance prompts
on optimal lighting and camera distance for accurate diagnosis.

02 Prescription OCR & Extraction

Scan. Extract. Checkout Faster.



Multimodal AI extracts
medication names, dosage strength, frequencies, and durations from handwritten doctor notes.

Auto-matches extracted drugs
against dark-store active inventories.

Pre-populates the 15-minute checkout cart
in a single click.

03 Emergency Red-Flag Detection

Detect. Alert. Save Lives.



Automatically identifies
critical trauma, acute burns, or severe allergic reactions.

Triggers 1-tap 108 Emergency Ambulance dispatch.

Pre-reserves emergency ICU/trauma beds at the nearest empaneled hospital.





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Breaking Language Barriers for Better Care

Multilingual Voice AI & Clinical Triage Engine

Real-time speech-to-text, auto-language detection, and structured clinical SOAP synthesis

Voice AI Technical Pipeline

From Patient Speech to Structured Clinical Intelligence

1. **Client-Side RMS Energy VAD**
   Web Audio API Voice Activity Detection detects voice pauses and auto-cuts audio transmission.
2. **Sarvam AI (saaras:v3)**
   16kHz mono speech recognition with real-time transcription in Telugu, Hindi, and English.
3. **Automatic Dialect Detection**
   Handles colloquial dialects like Tanglish ("naku jwaram undi") and Hinglish ("mujhe bukhar hai").
4. **Strict Target-Language Locking**
   Prompts enforce AI response strictly in the detected language with zero fallback drift.
5. **Sarvam Bulbul (bulbul:v3)**
   Regional HD female voice output (Kavitha, Kavya, Priya) for natural patient interaction.

AI-Generated SOAP Triage Notes

Structured, Actionable, and Multilingual-Ready

Patient: Lakshmi R.
Age: 32 years | **Gender:** Female
Language Detected: Telugu (with English terms)
Visit Time: May 16, 2025, 10:24 AM

Triage Priority: **Non-Urgent**

S — Subjective (S)
Patient-reported history

- Chief complaint: Fever and body pain
- Duration: 2 days
- Additional symptoms: Mild cough, headache
- Pain scale: 6/10
- Patient statement: "Naku jwaram undi, body pain ga undi."

O — Objective (O)
Extracted and inferred findings

- Temperature (reported): 102°F
- Other vitals: Not available (home assessment)
- Symptom severity: Moderate
- Red flags: None detected
- Associated symptoms: Mild cough, fatigue

A — Assessment (A)
Differential diagnosis (NVIDIA NIM - GPT-OSS-20B)

- Likely viral fever (e.g., influenza-like illness)
- Differential: Dengue (early), COVID-19 (mild), Seasonal viral URI
- Confidence: High
- Red flag assessment: No immediate emergency signs

P — Plan (P)
Actionable recommendations

- Advice: Rest, hydration, paracetamol 650 mg (if no allergies)
- OTC: ORS, vitamin C, steam inhalation
- Monitor: If fever > 3 days or breathing difficulty develops
- Referral: Consult a doctor if symptoms worsen

Any Language. Better Health. Together.

VOICE | INTELLIGENCE | BETTER CARE FOR EVERYONE

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Thank You

Questions & Discussion



